



Grade 8 SS Geography

1. Factors Affecting Temperature and Rainfall

Learning Guide

Introduction

What is Weather?

- **Weather** refers to the **daily atmospheric conditions** of a place.
- It includes things such as:
 - **Temperature**
 - **Rain**
 - **Wind**

What is Climate?

- **Climate** refers to the **average weather conditions over a long period**.
- This is usually measured over **30 years or more**.

Factor 1: Distance from the Equator / Latitude

Areas closer to the Equator are hotter because:

- The sun's rays hit the Earth **directly**.
- The heat is **concentrated on a smaller area**.
- These areas receive **more solar energy**.

Areas further from the Equator are cooler because:

- The sun's rays hit the Earth at a **slanted angle**.
- The heat is spread over a **larger area**.
- Less heat is received per **square metre**.

Example

- **Mbombela** is warmer because it is closer to the Equator.
- **George** is cooler because it is further from the Equator.

Factor 2: Distance from the Sea

Coastal Areas

Coastal areas are usually:

- **Cooler in summer**
- **Warmer in winter**

This happens because:

- The **sea heats and cools slowly**.
- The sea prevents **extreme temperatures**.
- The sea acts like an **air-conditioner**, helping to moderate temperatures.

Inland Areas

Inland areas are usually:

- **Hotter in summer**
- **Colder in winter**

This happens because:

- **Land heats and cools quickly**.
- Inland areas do not have the same **ocean influence**, so they experience greater temperature extremes.

Example

- **Cape Town** has more moderate temperatures.
 - **Johannesburg** has hot summers and cold winters.
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Factor 3: Height Above Sea Level / Altitude

High-altitude areas are cooler because:

- Air becomes **thinner** as height increases.
- Thin air cannot hold much heat.
- Temperature drops by about **1°C for every 100 m** increase in height.

Low-altitude areas are warmer because:

- The air is **denser** near the ground.
- Dense air traps more heat close to the surface.

Example

- **Johannesburg** is cooler because it is at a **high altitude**.
 - **Durban** is warmer because it is at a **low altitude**.
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Factor 4: Ocean Currents

Warm Ocean Currents

Warm ocean currents cause **higher temperatures** because:

- Warm water heats the **air above it**.
- Winds then blow this **warm air onto the land**.

Cold Ocean Currents

Cold ocean currents cause **lower temperatures** because:

- Cold water cools the **air above it**.
- Winds then blow this **cool air onto the land**.

Example

- **Durban** is warm because of the **Agulhas Current**.
 - **Port Nolloth** is cool because of the **Benguela Current**.
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Factor 5: Mountains / Relief

Windward Side: Cooler and Wetter

Mountains can make some areas **cooler and wetter** because:

- Air is forced to **rise**.
- As air rises, it **cools**.
- Cooling air causes **rainfall**.
- Higher ground is also **colder**.

Leeward Side: Hotter and Drier

Areas behind mountains are often **hotter and drier** because:

- Air sinks after crossing the mountain.
- As air sinks, it **warms**.
- This creates a **rain shadow effect**, leading to dry and hot conditions.

Example

- **Durban** receives more rain due to mountains.
- The **Karoo** is dry and hot because it lies in a **rain shadow** area.

Summary Table

Factor	Effect on Temperature / Rainfall	Example
Distance from Equator	Closer = hotter; further = cooler	Mbombela warmer, George cooler
Distance from Sea	Coastal areas have moderate temperatures; inland areas have extremes	Cape Town moderate, Johannesburg extreme
Altitude	Higher areas are cooler; lower areas are warmer	Johannesburg cooler, Durban warmer
Ocean Currents	Warm currents warm the coast; cold currents cool the coast	Agulhas warms Durban, Benguela cools Port Nolloth
Mountains / Relief	Windward side gets rain; leeward side is dry due to rain shadow	Durban wetter, Karoo drier

✓ Key Takeaways

- ✓ **Weather** is daily atmospheric condition.
- ✓ **Climate** is average weather over **30 years or more**.
- ✓ Areas closer to the **Equator** are warmer.
- ✓ The **sea** moderates temperature along the coast.
- ✓ Higher places are usually **cooler**.
- ✓ **Ocean currents** affect coastal temperatures.
- ✓ Mountains cause **rainfall on one side** and **dry conditions on the other side**.

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